

Task 16: Build a REST API Data Dashboard (Frontend Integration)

Tools:

- Primary: VS Code, Fetch API, JavaScript
- Alternatives: Axios (CDN), CodePen

Hints / Mini Guide:

1. Select a public REST API that returns structured JSON data such as users, products, or statistics.
2. Design a dashboard layout using HTML and CSS with sections for listing, details, and filters.
3. Use the Fetch API to retrieve data asynchronously from the REST endpoint.
4. Parse the JSON response and map the data into meaningful UI components.
5. Implement filtering and sorting features using JavaScript array methods.
6. Add pagination or lazy loading for better performance simulation.
7. Handle API errors gracefully and display user-friendly error messages.
8. Optimize DOM updates to avoid unnecessary re-renders.
9. Document the API endpoints used and explain data flow clearly.

Deliverables:

- Interactive data dashboard consuming REST API
- Real-world data handling and visualization logic

Interview Questions Related To Above Task:

- What is REST architecture?
- What are HTTP methods?
- How do you handle large datasets on frontend?
- What is pagination?
- Why is error handling important?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

